

without any bottlenecks. To be implemented using the IP protocol stack, 4G networks will provide connection speeds ranging from 100 Mbps to 1 Gbps, with assured security, high quality, and low cost. These networks will interoperate with existing networks and deliver applications like real-time streaming video, HDTV, and high-volume data.

5.2.5 CDMA

Code Division Multiple Access is an alternative technology to GSM for 2G and 3G networks. In CDMA, voice signals are digitised, and the frequency of the transmitted signal is varied according to a predefined code. This signal can only be intercepted by a receiver whose frequency response is programmed to vary with the same code. Since there are trillions of frequency-shifting sequences, CDMA signals offer considerable privacy. These signals use the 800 MHz and 1.9 GHz bands. CDMA signals utilise more bandwidth compared to GSM signals because of the variations in frequency. CDMA handsets are typically locked by carriers, which means you cannot switch your carrier without changing your handset.

5.2.6 CSD

Circuit Switched Data is a form of data transmission developed for TDMA networks like GSM. In this system, time slots are used to deliver data at the rate of 9.6 Kilobits per second. A newer variant is High Speed CSD (HSCSD) which employs more time slots and efficient coding methods to deliver data at enhanced rates.

5.2.7 EV-DO

Evolution-Data Optimised is a 3G protocol for wireless data transmission used mostly for broadband Internet access. EV-DO uses CDMA to transmit data at speeds of 500 to 1000 Kilobits per second, and is not compatible with GSM networks. Where signal strength is strong, EVDO enables a zone of pervasive computing where multiple devices can seamlessly use high-speed Internet access. Its primary competitor is HSDPA, which allows simultaneous voice and data transmission.